

A Multiband Passive Radar Demonstrator

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Abstract: *Passive radar systems that exploit signals from the plethora of RF emissions that exist in the external environment offer a number of advantages over conventional active radar system, including procurement and operational cost saving. Each emitter has its own characteristics, including waveforms, which dictate system performance. BAE Systems Advanced Technology Centre has designed and built a demonstrator system to act as a test bed for passive sensor research. The system operates over a multi-octave bandwidth and can exploit both analogue and digital transmissions from both broadcast and communication systems. This offers advantages over single band systems; for example, multiple observations in different bands using different geometries allow fusion of tracks to achieve more robust and accurate tracks.*

This paper outlines the system and design issues that were addressed during the development of the demonstrator, including the simulation model, and prediction of the bistatic target signature. Results of experimental work illustrating the operation of the demonstrator system with targets of opportunity are shown.

1. Introduction

Passive coherent location (PCL) systems with the capability to routinely detect and track air targets using ‘transmitters of opportunity’ have been widely reported [2]. These systems exploit signals from the many RF emissions from diverse sources and offer a number of advantages over conventional active radar systems, including procurement and operational cost savings from using these third party transmissions, lower power requirements and covertness. Typically such systems exploit RF emissions produced by communications, radio or television broadcast services, where each emitter has its own characteristics in terms of azimuth and elevation coverage, power level and waveform.

Following work on forward scatter radar using analogue television transmissions [1], BAE Systems Advanced Technology Centre has designed and built a demonstrator system to act as a test bed for passive sensor research. The demonstrator system is designed to operate over a multi-octave bandwidth and can be configured to exploit both analogue and digital transmissions from both broadcast and communication systems. This maximises flexibility, and allows surveillance and tracking by exploiting the optimum radar returns based on the particular geometry, coverage, waveform and target signature.

This paper describes the demonstrator system, the simulation model used to verify the concept, prediction of bistatic target signatures, design and system issues that were addressed during the development of the demonstrator, and finally presentation of some results of experimental work, illustrating the operation of the demonstrator system with targets of opportunity.

2. PCL Simulation Model

To verify the PCL system concept a simulation model was developed which represents the PCL demonstrator. This built on work performed in-house in developing a Forward Scatter system [1]. The application, developed in MATLAB, encompasses the synthesis of signals from aircraft responses to transmitters typical of analogue and digital TV and radio

transmissions, and the digital signal processing of the system. Of particular interest is the COFDM family of waveforms transmitted by DAB and DVB-T transmitters.

The signal processing chain includes the digital down-conversion to baseband of the digitised signals recorded by a high-speed capture card, polyphase decimation to perform low-pass filtering to filter out out-of-band signals and to decimate, weighted beamforming, adaptive digital null-steering, cross ambiguity processing and detection processing.

An adaptive null-steering stage uses a reference beam formed in the direction of the transmitter of interest. The sequential decorrelation algorithm is used to derive weights which steer a null in the beams used for target detection.

The cross-ambiguity processing provides estimates of the time-difference of arrival (TDOA) of target returns with respect to the direct signal from the transmitter (lag), and the Doppler of those targets. It is estimated as follow:

$$\chi_{lag,Doppler} = \sum_{t=1}^{N_{samples}} wfft_t \cdot ref_{t-lag}^* \cdot sig_t \cdot e^{-2\pi j \cdot Doppler \cdot t} \quad (1)$$

where $wfft$ are the window weights for sidelobe reduction
 ref is the reference signal, from a reference beam
 sig is the signal from the beam of interest

The detection processing consists of fixed or adaptive (CFAR) thresholding applied to the cross ambiguity function, providing detections with estimated lag and Doppler values.

Simulation study showed that with realistic power levels representative of transmitter and bistatic target signals (see next Section) targets are detectable with the processing assumed. The study emphasized the need for null-steering to produce detections.

3. Target Signatures

To ascertain the bistatic RCS of a typical target, required as input in the simulation model, the bistatic signatures for several different types of aircraft were derived for selected RF bands. For small single-seater aircraft, such as the Piper Cherokee, detailed electromagnetic simulation using an in-house developed finite difference time domain code (FDTD) was used. For larger aircraft, e.g. Boeing 737, a prohibitive amount of computer time would be required for rigorous prediction of the RCS at the frequencies of interest. Instead, an acceptable approximation to their RCS can be obtained with a physical optics prediction of the RCS of a cylinder to represent the fuselage. Although this neglects the contribution of the wings and tail fin to the RCS, it provides a first order approximation for the system calculations.

In summary, the maximum RCS of a Piper Cherokee is of the order of 30 dBm² at 220MHz and 38dBm² at 650MHz, but in general does not exceed 0 to 5dBm² and is lower at higher RF. For a Boeing 737 the maximum RCS is 39dBm² at 220MHz and 43dBm² at 650MHz, but averages out at between 10 to 20dBm².

4. Demonstrator Design

The rationale for the demonstrator design was to develop a system capable of receiving and processing analogue and digital transmissions over a band covering 200MHz to 2GHz and one which was scalable and low-cost. A four-channel system was developed capable of capturing data at 10MHz. This provides sufficient bandwidth to process the analogue and digital transmissions of interest but also permits sustained data capture. The wider bandwidth of digital transmissions provides for good range resolution.

The antenna system consists of an array of four wideband Log-Periodic Dipole antennas. Each antenna has a 200MHz to 2GHz bandwidth. The antennas are at a 45° angle from vertical to permit joint reception of vertically and horizontally polarised signals, with some loss. The antennas are spaced so that there are no grating lobes at the highest RF of interest.

The receiver was designed using relatively inexpensive off-the-shelf components. This led to rapid, low-risk implementation of a working system. The main stages of the receiver include multiple IF stages with appropriate filtering. The IF frequencies were chosen so that the receiver was able to process signals in the desired RF range.

For data acquisition on a PC an ICS-645-4A 32-bit 33MHz PCI card was selected. This provides four onboard ADCs clocked at up to 20MHz. The PCI 32/33 standard allows for a maximum burst data rate of 132MB/sec., however this is not a sustained rate. Bus sharing and other factors reduce this rate significantly. Initial measurements with the PC and using in-house data capture software resulted in a sustained rate of nearly 50MB/sec. The required data rate for the PCL system is 80MB/sec, since it is sampling four channels at two bytes per channel and at 10MHz rate. A capture rate less than this results in data loss. A system upgrade, which included a RAID 0 array with dedicated raid controller card, multiple PCI buses, and enhanced data capture software enabled the required 80MB/sec rate to be obtained.

The main processing stages are described in Section 2. For the demonstrator, channel calibration is additionally required. This stage is performed in software prior to beamforming, and corrects the amplitude and phases of each channel to account for mismatches in the components in the system. The appropriate weighting for each channel is determined using a known transmitter as a reference. The predicted amplitude and phase response of the system to the reference is calculated and used to correct the actual amplitudes and phases of each channel. The complete processing software has been designed efficiently to provide near real-time processing of received signals.

5. System and Design Issues

A number of system and design issues were addressed during the programme, leading to iterations in the system design and providing direction for future enhancements to the system. The major issue, which is an issue with all passive sensing systems, is that of receiver dynamic range. High dynamic range, in excess of 90 to 100dB, is required to accommodate the large signals from the transmitters used and allow detection of the weaker bistatic scatter from targets. Bistatic target scatter is typically small, as indicated in Section 3, unless the target is in the forward-scatter region, where its RCS will be enhanced. In the current system, filtering of the direct transmitter signal is provided by an adaptive digital null-steering algorithm. This has been shown to provide some transmitter cancellation but insufficient to provide reliable detection. Analogue suppression of the direct signal at the element level to reduce the ADC dynamic range requirement is one option that is being addressed.

With a system that operates with wideband signals there is an issue with the response of components, such as filters, over such bandwidths. To optimise performance, receiver channel responses must be matched and this requires some form of calibration, e.g. channel equalisation filters.

6. Initial Results

Results illustrating the operation of the demonstrator system with targets of opportunity are presented here. The demonstrator system was sited at BAE Systems ATC in Great Baddow, UK. There are a number of transmitters within the coverage of the system. It was decided to use the DVB-T transmissions from the transmitter at Sudbury, 38km away. With London

Stansted airport close by this baseline provides good opportunity to detect aircraft landing and departing from the airport on a frequent basis. Signal calculations were made to determine that detections were indeed possible with the system using this geometry. An ADS-B receiver, connected to a PC, was used to cue data recording by allowing aircraft positions in the PCL coverage to be visualised. The following figures are examples of the cross-ambiguity map obtained from measurements.

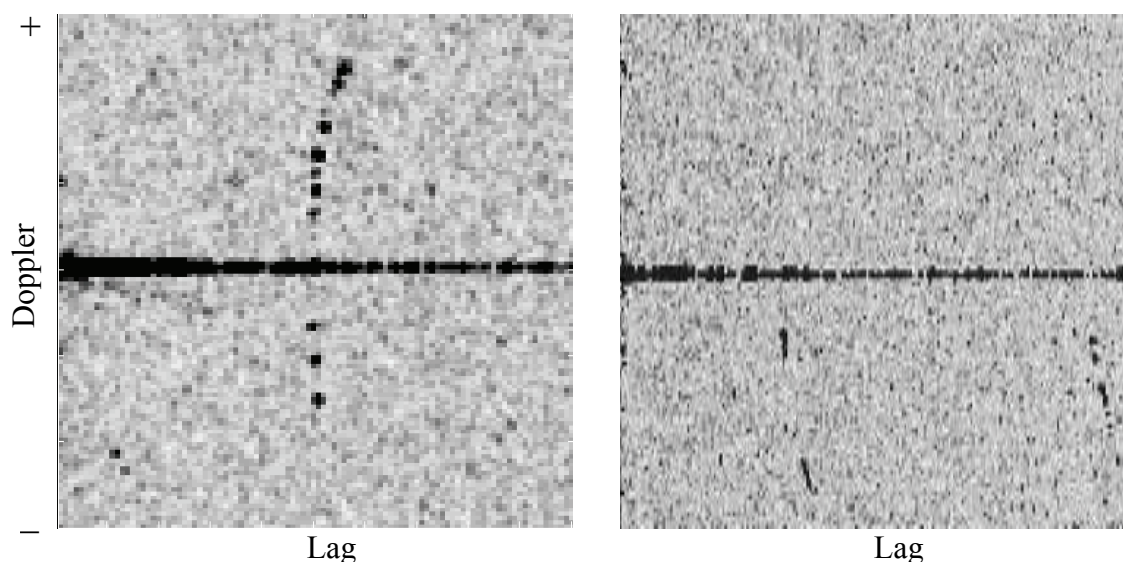


Figure 1. Examples of Target Detections obtained with DVB-T transmissions

These were shown, via ADS-B data, to be consistent with aircraft approaching Stansted and those in the holding stack for the same airport. The large returns at zero Doppler are the auto ambiguity returns of the transmitter signal which have been partially cancelled by the digital null-steering algorithm, and further work to enhance the performance is in progress.

7. Summary

Using off-the-shelf technology a low-cost demonstrator system was designed and implemented to receive and process multi-band analogue and digital transmissions. Multi-band passive sensing provides a robust, inexpensive solution for locating targets. The concept has been demonstrated using commercial digital transmissions. A further iteration of the system is currently in progress, providing analogue null-steering and automatic RF switching capability, and this should provide a robust, true multi-band capability for demonstration in further planned trials in 2006.

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